Ariyeh Weissman-Bennett

Mechanical Engineering | University of Massachusetts Amherst

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EDUCATION

University of Massachusetts Amherst | B.S. Mechanical Engineering | Expected May 2027

Coursework: Advanced Fluid Mechanics, Aerospace Structures, Thermodynamics, Design of Mechanical Components & Assemblies, Dynamic Systems, Heat Transfer, Electrical Engineering, Manufacturing Processes.

TECHNICAL SKILLS

SOLIDWORKS, CAD, Ansys, MATLAB, Simulink, Python, Arduino, FDM 3D printing, manual machining, mechanical design, DFMA, GD&T.

ENGINEERING EXPERIENCE

Undergraduate Research Intern | THEAM Lab, UMass Amherst
Aug 2026 - Present

• Prepared pNIPAM hydrogel resin and fabricated custom chips for two-photon direct laser writing experiments.
• Fabricated hydrogel microstructures and measured thermoresponsive behavior.
• Investigated size, structure, and material effects on response time and microactuator performance.

Machine Learning & Code Lead | Electric Motor Fault Detection Capstone
Sep 2026 - Present

• Developing and validating a Simulink electric motor model for normal and faulty operating conditions.
• Applying MATLAB signal processing and feature extraction for machine-learning model development.
• Integrating vibration sensors and a microcontroller for signal collection and model validation.

Teaching Assistant | Strength of Materials & Additive Manufacturing
UMass Amherst | Jan 2026 - Present

• Held office hours on strength of materials and mechanical design.
• Supported CAD preparation, slicing, printer setup, and FDM troubleshooting.

SELECTED DESIGN PROJECTS

Lightweight Wrench | Team Member | Sep - Dec 2025

Designed and optimized an aluminum wrench in SOLIDWORKS. Performed hand stress analysis and Ansys static structural FEA. Machined the final part and validated performance through three-point bending and functional torque testing.

Mini Smart Car | Individual Contributor | Dec 2025

Built an Arduino Uno vehicle using breadboarded circuitry, DC motors, and sensors. Programmed obstacle avoidance and LED status indicators.

HONORS

Dean’s List; Carl Christenson UMass Engineering Scholarship; Henry ‘Hank’ J. Picard UMass Engineering Scholarship; Somerville Mathematics Fund Scholarship.